



AI Contract Risk Analyzer

ClearClause

TURNING DENSE LEGAL TEXT INTO 60-SECOND DECISIONS.

A hybrid NLP system that segments freelance and professional contracts, detects 17 categories of risky clauses, scores the document, benchmarks each clause against a fair-contract standard, and coaches the user through the specific negotiation moves to make before signing.

PROJECT GROUP

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Contracts are written for lawyers, not freelancers.

Most freelancers and small businesses sign multi-page agreements without fully understanding what they're agreeing to. Risky clauses around IP, termination, liability and exclusivity routinely cost the smaller party more than the project fee itself.

~4,800

words in the typical freelance contract analysed by ClearClause

17

clause categories tracked, from IP and indemnity to auto-renewal

96/100

max risk score the highest-risk demo earns before suggested edits

"Even when the words are clear, the consequences are not. A small wording change in section 9 is often the difference between getting paid in 15 days and waiting 90."

ClearClause — a hybrid-NLP contract review copilot.

Upload a contract; ClearClause segments, classifies, scores, explains, benchmarks, answers questions, and exports — entirely offline. An optional Groq layer polishes wording.

Detect

- 17 clause categories
- Regex catalog + TF-IDF semantic similarity
- Every match is traceable to evidence

Score

- 0–100 risk score with verdict card
- Severity adjustment via red / green flags
- Exponential curve avoids saturation

Coach

- Priority-sorted negotiation checklist
- Fair-contract benchmark per clause
- Suggested discovery questions

Export

- Presentation-quality HTML report
- Clause CSV + Markdown pack
- Document-grounded Q&A chat

The NLP pipeline at a glance.

Each step has a single responsibility and a clean dataclass interface, so the whole pipeline is explainable end-to-end.

1 Parse

TXT / PDF / DOCX → cleaned text + paragraphs.

2 Segment

Block-level section splitter (\Z-anchored regex).

3 Extract

Regex patterns + TF-IDF semantic similarity + flag counts
→ confidence.

4 Score

Confidence-weighted weights, severity bonus/discount,
exponential curve.

5 Benchmark

Compare each clause to fair-standard text; flag missing
core clauses.

6 Coach + Q&A

Priority checklist, suggested questions, TF-IDF retrieval
Q&A.

ParsedDocument → Section[] → ClauseFinding[] → RiskProfile → AnalysisResult → HTML / CSV / Markdown

17 clause categories grouped into 5 risk dimensions.

Money & Liability

Payment Terms

Unlimited Liability

Indemnification

Limitation of Liability

Ownership & IP

IP Ownership / Work-for-Hire

Confidentiality / NDA

Data Privacy / Security

Termination & Renewal

Unilateral Termination

Automatic Renewal

Work Freedom

Non-Compete

Exclusivity

Non-Solicitation

Disputes

Governing Law & Jurisdiction

Arbitration / Court Waiver

Operations

Scope, Acceptance & Revisions

Assignment / Change of Control

Effective Date & Term

Three signals combine into one explainable confidence score.

1 • Regex catalog match

Hand-curated patterns and red/green flag terms per clause.
Captures the explicit, traceable signal.
Example: \birrevocably assigns?\b → IP Ownership.

2 • TF-IDF semantic similarity

Cosine similarity between each section and the rule's prototype text.
Catches paraphrased or unusual wording the regex misses.
Adds a continuous signal alongside binary pattern matches.

3 • Red / green flag counters

Red flags (e.g. 'unlimited', 'sole discretion') push severity up.
Green/balancing flags (e.g. 'fees paid', 'mutual') pull severity down.
Drives the per-clause severity adjustment.

WORKED EXAMPLE

IP Ownership clause in the high-risk freelance demo

"To the extent any work product is not work-for-hire, Contractor irrevocably assigns all right, title, and interest in such intellectual property to Client. Client shall exclusively own all deliverables."

Regex pattern matches	intellectual property · irrevocably assigns · all right, title, interest
Red flag terms	irrevocably · exclusively owned · all right
Green / balancing terms	(none)
TF-IDF similarity to prototype	0.42
Final confidence	0.98
Adjusted severity	HIGH (no demotion — no balancing language)

Transparent formula. No saturation at the top.

Formula

```

contribution = (weight + red_bonus - green_discount) ×
conf_factor
raw_total = Σ contributions + 3.5·high_count +
1.25·medium_count
if raw_total ≤ 60:    score = round(raw_total)
else:                score = round(60 + 36·(1 - exp(-
(raw_total - 60) / 160)))
                    ⇒ asymptote ≈ 96, ceiling = 100

```

Verdict thresholds

≥ 78	Critical Risk	Do not sign in current form.
60 – 77	High Risk	Revisions required before signing.
30 – 59	Review Carefully	Manageable; clarify wording.
< 30	Low Risk	Final read-through recommended.

Score saturation, before vs. after

Before the curve, two very different bad contracts both hit 100. After the curve they spread out — the dashboard finally has range.



Spread now 90 → 85 → 80 → 53 → 25 across all 5 demos.

From detection to action — what to ask for.

#1 • Indemnification

HIGH • 98%

Legal-defense costs can be much higher than the project value.

- ✓ Make indemnity mutual: each party covers claims it actually caused
- ✓ Limit the trigger to 'to the extent caused by' your own breach, negligence, or IP claim
- ✓ Require the client to control defense and let you approve any settlement

Fair-contract benchmark

Each detected clause is compared to a fair-standard version. Missing standard clauses are flagged too.

Clause	Status	Detected language vs. fair standard
IP Ownership	high-risk	Irrevocable assignment with no carve-outs.
Unilateral Termination	high-risk	Sole-discretion termination, no kill fee.
Liability Cap	concerning	One-sided cap protects only the client.
Confidentiality	fair	Mutual, 3-year term, standard exceptions.
Non-Solicitation	missing	Standard clause not present — ask why.

Full contract with risky spans highlighted in place.

The Streamlit redline view stitches all detected spans back into the source text so a reviewer can see at a glance which paragraphs need attention — hover any highlight for the clause name.

HIGH • 7

MEDIUM • 7

LOW • 2

(from the high-risk freelance demo)

3. Intellectual Property

Contractor irrevocably assigns all right, title, and interest to Client.

5. Non-Compete

Contractor shall not compete with Client for 24 months after termination.

8. Termination

Client may terminate this Agreement at its convenience upon 7 days notice.

12. Governing Law

Governed by the laws of Delaware; exclusive jurisdiction in New Castle County.

Grounded retrieval Q&A — no hallucinations.

Retrieval flow

1

Chunk contract

Split into ~950-char overlapping paragraphs.

2

Match clause intent

Keyword map for 17 topics → shortcut to detected clause.

3

TF-IDF retrieve

Cosine top-k chunks for the user's question.

4

Respond

Use the clause's plain-English answer + cite contract evidence.

Example exchange

Who owns the final source code?

Detected: IP Ownership (HIGH • 98% confidence)

Plain English: The contract assigns all IP, source code, and deliverables to the client. You will not own the finished work.

Negotiation note: Carve out pre-existing tools, templates, and generic libraries; reserve portfolio display rights.

Evidence: "Contractor irrevocably assigns all right, title, and interest in such intellectual property to Client. Client shall exclusively own all deliverables."

Score, verdict, radar, KPI strip, and ranked red flags.

Risk score

90/100

High-risk clauses

7

Medium-risk clauses

7

Low + Info clauses

3

Avg confidence

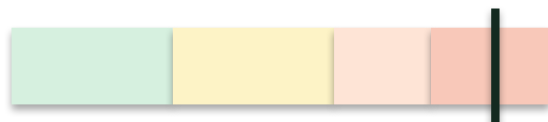
89%

CLEARCLAUSE VERDICT

Critical Risk — Do not sign in the current form.

Concentrates several high-severity clauses against one party. Push for revisions on the priority red flags before signing.

Composite risk score



90/100

Critical Risk

Risk by business area



Priority red flags

Indemnification

Legal-defense costs can exceed project value.

IP Ownership

Loss of source-code and portfolio rights.

Non-Compete

Blocks main source of income after contract.

Unlimited Liability

Personal exposure beyond fee.

Gold (in-domain) + held-out (out-of-domain) = honest metrics.

The hand-curated gold set scores 1.00 across the board, but that's the optimistic number because the catalog was tuned against this contract. The held-out demo is from a completely different domain (influencer marketing) and reports the realistic number.

Gold demo · high-risk freelance contract

17 expected clause IDs · catalog was tuned against this contract

Precision

1.00

Recall

1.00

F1

1.00

TP: 17 · FP: 0 · FN: 0

Every expected clause type is detected. Confidence is high (avg 0.87), and severity counts (7 HIGH / 8 MEDIUM) match a sensible legal review.

Held-out · creator / influencer agreement

11 expected clause IDs · catalog was NOT tuned for this domain

Precision

0.92

Recall

1.00

F1

0.96

TP: 11 · FP: 1 (non_compete) · FN: 0

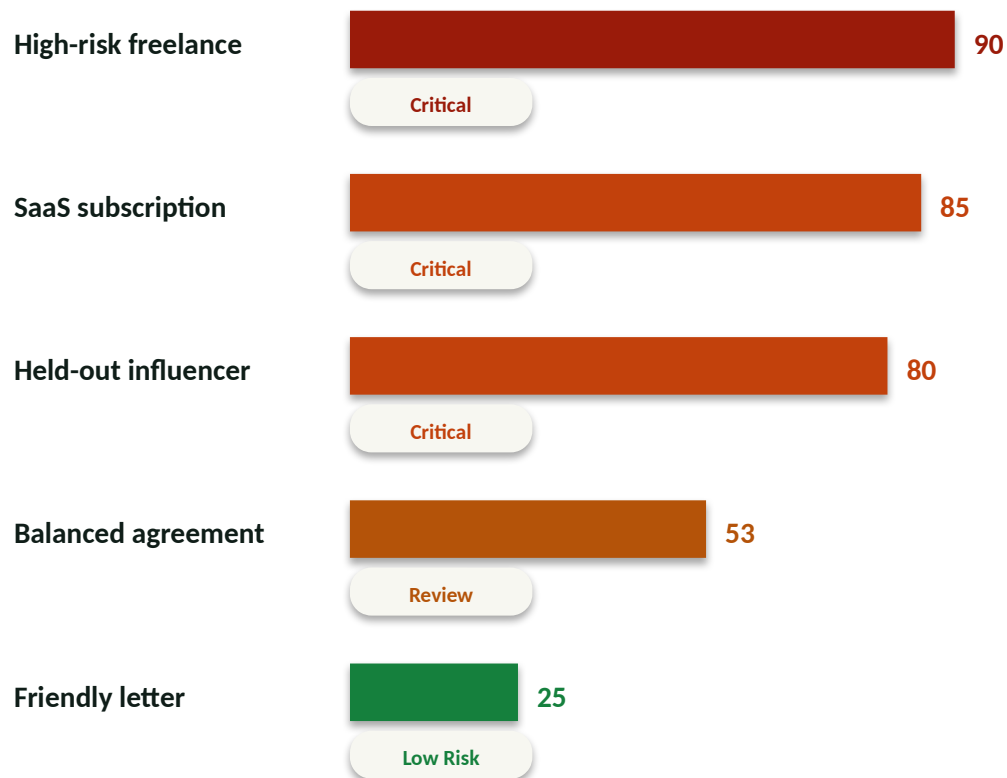
The one false positive (non_compete) flags a section that is really 'Category Exclusivity'. Honest failure mode worth discussing in viva — a real limit of the catalog on creator contracts.

21 / 21 unit tests pass · pipeline, exports, Q&A, score spread, held-out evaluation, calibration, redline rendering.

The model is well-calibrated and the gauge has real range.

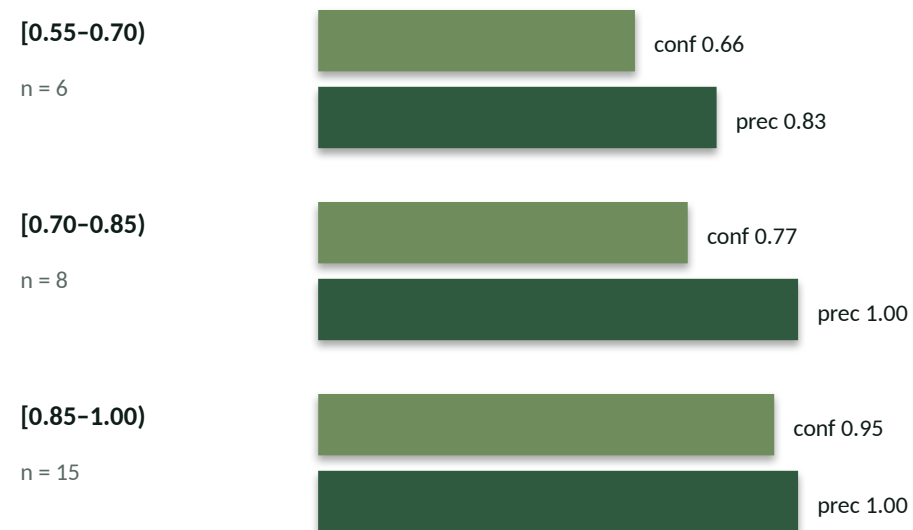
Risk score across the 5 bundled demos

Same pipeline, no per-contract tuning. Each demo lands in a distinct band.



Confidence calibration

Mean confidence vs. empirical precision per bin, across 29 gold + held-out findings.



Brier = 0.053 • ECE = 0.126 • overall precision = 0.97

Model is slightly under-confident: when it says 0.95 it is right 100% of the time.

What's in the box — and what we'd build next.

01 / TECH STACK

Reproducible, lightweight

- Streamlit · interactive UI with custom CSS
- scikit-learn · TF-IDF + cosine similarity
- Plotly · gauge, donut, radar, calibration chart
- pdfplumber + python-docx · document parsing
- Groq · optional LLM polish (works offline by default)

02 / KNOWN LIMITS

Honest weak spots

- Hand-curated catalog → recall capped to known clauses
- English only (no Urdu / Arabic support yet)
- Section splitter assumes paragraph-block formatting
- False positives possible when fair text mentions risky terms
- No version-comparison or diff view

03 / WHAT'S NEXT

If we kept building

- Sentence-Transformer classifier trained on CUAD
- Inline contract diff between two versions
- Multi-language support (Urdu + Arabic clauses)
- Real PDF export with WeasyPrint
- Cloud deployment + share-link mode for clients

Thank you

Open the Streamlit app and walk the examiner through the high-risk demo,
then switch to the friendly demo to show the gauge in green.

REPOSITORY

github.com/ShehrozRafaqat/ClearClause-NLP-Project

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Streamlit • scikit-learn • Plotly • optional Groq